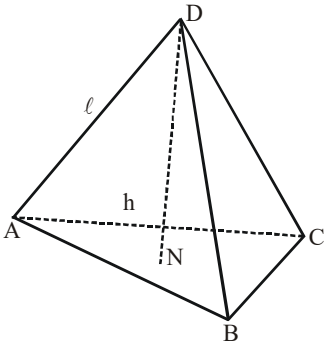
SOLUTION

SECTION–I

1. Ans. (A)

Sol. Let D is origin and position vectors of A, B, C are $\vec{a}, \vec{b}, \vec{c}$ respectively,



then $|\vec{a}| = |\vec{b}| = |\vec{c}| = \ell$ and

$$|\vec{a}.\vec{b}| = |\vec{b}.\vec{c}| = |\vec{c}.\vec{a}| = \ell^2 \cos 60° = \frac{\ell^2}{2}$$

Now $$[\vec{a}\,\vec{b}\,\vec{c}]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{c}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} = \frac{\ell^6}{2}$$

now volume of tetrahedron $= \frac{1}{3} \times$ Area of base $\times$ height

$$\Rightarrow \quad \frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}] = \frac{1}{3} \times \left(\frac{\sqrt{3}}{4}\ell^2\right) \times \text{height}$$

$$\Rightarrow \quad \text{hight} = \frac{\frac{\ell^3}{\sqrt{2}} \times \frac{1}{2}}{\frac{\sqrt{3}}{4}\ell^2} = \sqrt{\frac{2}{3}}\ell$$

2. Ans. (D)

Sol. Lines L_1 & L_2 are coplanar if

$$\begin{vmatrix} 3-2 & 2-3 & 1-2 \\ 2 & 3 & \lambda \\ 3 & 2 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

Let $L_1 : \frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{4} = t$

$\therefore$ Any point on it is P(2t + 3, 3t +2, 4t +1)

It lies on the plane x + y + z = 15 $\Rightarrow$ 9t + 6 = 15 or t = 1

$\therefore$ Distance of P(5, 5, 5) from (3, 4, 3) = 3

3. Ans. (D)

Sol. $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin 30° \hat{n} = 3\hat{n}$

Since $\vec{d}$ is perpendicular to both $\vec{a}$ & $\vec{b}$

$\Rightarrow \quad \hat{n} = \pm\vec{d}$

$\therefore \quad \vec{a} \times \vec{b} = \pm 3\vec{d}$

Now $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \pm 3\vec{d} \times (\vec{c} \times \vec{d}) \Rightarrow$

$$\pm 3\{(\vec{d}.\vec{d})\vec{c} - (\vec{d}.\vec{c})\vec{d}\} = \frac{\hat{i}}{6} - \frac{\hat{j}}{3} + \frac{\hat{k}}{3} \quad \text{or}$$

$$\pm 3\vec{c} = \frac{\hat{i} - 2\hat{j} + 2\hat{k}}{6}$$

$$\therefore \quad |\vec{c}.\hat{i}| = \frac{1}{18}, |\vec{c}.\hat{j}| = \frac{1}{9}, |\vec{c}.\hat{k}| = \frac{1}{9}$$

4. Ans. (C)

Sol. Diagonals of rhombus are perpendicular

$\Rightarrow \quad \vec{d}_1.\vec{d}_2 = 0 \quad \Rightarrow \quad 2|\vec{a}|^2 - 2|\vec{b}|^2 + 3\vec{a}.\vec{b} = 0$ ....(i)

& $\quad \vec{d}_3.\vec{d}_4 = 0 \quad \Rightarrow \quad 2|\vec{a}|^2 - 2|\vec{b}|^2 - 3\vec{a}.\vec{b} = 0$ ....(ii)

$\Rightarrow \quad \vec{a}.\vec{b} = 0 \quad \Rightarrow \quad \vec{a}$ & $\vec{b}$ are perpendicular

$\Rightarrow$

$$[\vec{a}\ \vec{b}\ \vec{a} \times 2\vec{b}] = (\vec{a} \times \vec{b}).(\vec{a} \times 2\vec{b}) = 2|\vec{a} \times \vec{b}|^2 = 2|\vec{a}|^2|\vec{b}|^2$$

5. Ans. (D)

Sol. $\vec{c} = \alpha\left(\hat{i} + \hat{j}\right) + \beta\left(\hat{j} + \hat{k}\right)$

$$= \alpha\hat{i} + (\alpha + \beta)\hat{j} + \beta\hat{k}$$

$\because$ $\vec{c}, \hat{i} - 2\hat{j} + \hat{k}, 3\hat{i} + 2\hat{j} - \hat{k}$ are coplanar

$$\therefore \begin{vmatrix} \alpha & \alpha+\beta & \beta \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{vmatrix} = 0$$

$\alpha(2 - 2) - (\alpha + \beta)(-1 - 3) + \beta(2 + 6) = 0$

$4\alpha + 4\beta + 8\beta = 0$

$$\frac{\alpha}{\beta} = -3$$

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6. Ans. B,C

Sol. Let equation of plane be $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

it cuts coordinates axes at A(a,0,0), B(0,b,0), C(0,0,c)

its centroid is $G\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)\equiv\left(1,r,r^2\right)$

$\Rightarrow a=3,\ b=3r,\ c=3r^2$

$\Rightarrow$ equation of plane is $x+\frac{y}{r}+\frac{z}{r^2}=3$

$\because$ it posses through (4,–8,–15)

$\therefore\ 4-\frac{8}{r}+\frac{15}{r^2}=3\Rightarrow r^2-8r+15=0$

$\Rightarrow r=5, r=3$

7. Ans. B,D

Sol. Let normal vector n_1 perpendicular to plane determining $\hat{i},\hat{j}+\hat{k}$ is

$n_1=\hat{i}\times(\hat{i}+\hat{j})=\hat{k}$

similarly $n_2=(\hat{i}-\hat{j})\times(\hat{i}-\hat{k})=\hat{i}+\hat{j}+\hat{k}$

Now vector parallel to intersection of plane $=\vec{n}_2\times\vec{n}_1$

$=\vec{k}\times(\hat{i}+\hat{j}+\hat{k})=-(\hat{j}-\hat{i})\ \Rightarrow\ \frac{x-0}{1-0}=\frac{y-0}{1-0}=\frac{z-0}{1-0}$

Angle between $\lambda(-\hat{j}+\hat{i})$ and $(\hat{i}-2\hat{j}+2\hat{k})$

$\cos\theta=\frac{\lambda(-\hat{j}+\hat{i}).(\hat{i}-2\hat{j}+2\hat{k})}{\lambda\sqrt{2}\times3}=\frac{1}{\sqrt{2}}\ \Rightarrow\ \theta=\frac{\pi}{4}$ or θ

$=\pi-\frac{\pi}{4}=\frac{3\pi}{4}$

8. Ans. (A,B,C)

Sol. We have $\hat{i}.\hat{j}=0$

given $\vec{p}=3\hat{i}+4\hat{j},\ \vec{q}=5\hat{i}$

$4\vec{r}=\vec{p}+\vec{q},\ 2\vec{s}=\vec{p}-\vec{q}$

$\Rightarrow\ \vec{r}=2\hat{i}+\hat{j},\ \vec{s}=-\hat{i}+2\hat{j}$

for option A :

$|\vec{r}+k\vec{s}|=\left|(2-k)\hat{i}+(1+2k)\hat{j}\right|$

$=\sqrt{(2-k)^2+(1+2k)^2}$

$=\sqrt{5+5k^2}$

..... (i)

& $|\vec{r}-k\vec{s}|=\left|(2+k)\hat{i}+(1-2k)\hat{j}\right|$

$=\sqrt{(2+k)^2+(1-2k)^2}$

$=\sqrt{(5+5k^2)}$

..... (ii)

from (i) and (ii)

$|\vec{r}+k\vec{s}|=|\vec{r}-k\vec{s}|$

option B :

$\because\ \vec{r}.\vec{s}=0$

$\vec{r}\perp\vec{s}$

option C :

$\because\ \vec{r}+\vec{s}=\hat{i}+3\hat{j}$ and $\vec{r}-\vec{s}=3\hat{i}-\hat{j}$

$(\vec{r}+\vec{s}).(\vec{r}-\vec{s})=0$

Hence $(\vec{r}+\vec{s})$ is perpendicular to $(\vec{r}-\vec{s})$

option D :

$|\vec{r}|=\sqrt{5}\ ,\ |\vec{s}|=\sqrt{5}\ ,\ |\vec{p}|=5\ ,\ |\vec{q}|=5$

$\therefore\ |\vec{r}|=|\vec{s}|\neq|\vec{p}|=|\vec{q}|$

9. Ans. (A,C)

Sol. The required vector, $\vec{r}=\lambda\left(\vec{a}+\vec{b}\right)$, λ is a scalar or

$\vec{r}=\lambda\left(\frac{1}{9}\left(7\hat{i}-4\hat{j}-4\hat{k}\right)+\frac{1}{3}\left(-2\hat{i}-\hat{j}+2\hat{k}\right)\right)$

$=\frac{\lambda}{9}\left(\hat{i}-7\hat{j}+2\hat{k}\right)$

$\because\ |\vec{r}|^2=54\ \Rightarrow\ \frac{\lambda^2}{81}(1+49+4)=54$

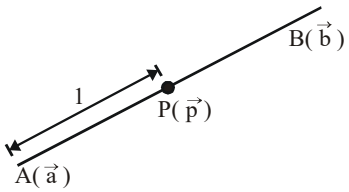
$\Rightarrow\ \lambda=\pm9$

10. Ans. (A,D)

Sol. $\vec{a}=\hat{i}+\hat{j}-2\hat{k},\ \vec{b}=\hat{i}-3\hat{j}+\hat{k}$

$\vec{p}-\vec{a}=\lambda(\vec{b}-\vec{a})=\lambda(-4\hat{j}+3\hat{k})$

$|\vec{p}-\vec{a}|=\lambda\sqrt{16+9}$



$25\lambda^2=1\Rightarrow\lambda=\pm\frac{1}{5}$

$\therefore\ \vec{p}=\vec{a}+\lambda(\vec{b}-\vec{a})=\begin{bmatrix}\frac{1}{5}(5\hat{i}+\hat{j}-7\hat{k})\\ \frac{1}{5}(5\hat{i}+9\hat{j}-13\hat{k})\end{bmatrix}$

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Paragraph for Question 11 to 13

Sol. Plane containing the line

$L_1 : 2x - y + 3z - 5 = 0 = x - 2y + 2z + 1$

is in the from $P_1 + \lambda P_2 = 0$

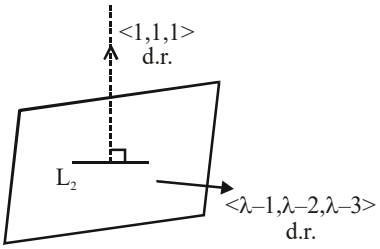
where $P_1 : 2x - y + 3z - 5 = 0$, $P_2 : x - 2y + 2z + 1 = 0$

$(2x - y + 3z - 5) + k(x - 2y + 2z + 1) = 0$

it passes through (1,2,3) [$\because$ plane contains lie L_2]

$(2 - 2 + 9 - 5) + k(1 - 4 + 6 + 1) = 0$

$4 + k(4) = 0 \Rightarrow k = -1$



11. Ans. (A)

Sol. Equation of plane $x + y + z = 6$

12. Ans. (C)

Sol. L_2 also lies on the plane

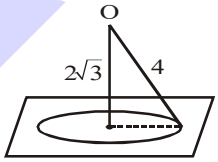
$(\lambda - 1) \cdot 1 + (\lambda - 2) \cdot 1 + (\lambda - 3) \cdot 1 = 0$

$\Rightarrow \lambda = 2$

13. Ans. (C)

Sol. Perpendicular distance of (0, 0, 0) from the plane

$\Rightarrow \frac{6}{\sqrt{3}} = 2\sqrt{3}$



Hence radius of circle be 2.

$\Rightarrow$ required area is 4π

SECTION–II

1. Ans. 3.96

Sol. $\because \left(\vec{a} \times (\vec{a} \times \vec{b})\right) \times \vec{c} = \left((\vec{a}.\vec{b})\vec{a} - (\vec{a}.\vec{a})\vec{b}\right) \times \vec{c}$

$= -|\vec{a}|^2 \vec{b} \times \vec{c} = -6(\vec{b} \times \vec{c})$

Similarly $\left(\vec{b} \times (\vec{b} \times \vec{c})\right) \times \vec{a} = -|\vec{b}|^2 \vec{c} \times \vec{a}$

$= -2(\vec{c} \times \vec{a})$

and $(\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b} = -|\vec{c}|^2 \vec{a} \times \vec{b} = -3(\vec{a} \times \vec{b})$

Now, $\vec{v} = -\left[3(\vec{a} \times \vec{b}) + 6(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})\right]$

$\because$ $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular

$\therefore$ $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$ and $\vec{c} \times \vec{a}$ are also mutually perpendicular

$\Rightarrow$

$|\vec{v}| = \sqrt{9|\vec{a}|^2|\vec{b}|^2 + 36|\vec{b}|^2|\vec{c}|^2 + 4|\vec{c}|^2|\vec{a}|^2} = 6\sqrt{11}$

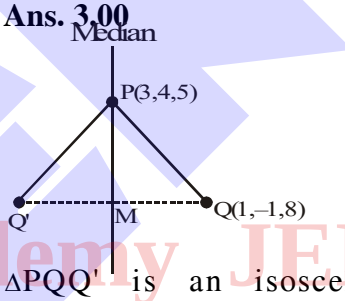
$M = 6\sqrt{11}$

$M^2 = 36 \times 11$

$\frac{M^2}{100} = 3.96$

2. Ans. 3.00

Sol.



$\Delta PQQ'$ is an isosceles triangle & $\overrightarrow{PM} = 2\hat{i} - 2\hat{j} + \hat{k}$

Now, $PM = |\overrightarrow{PQ}|\cos\theta$, $\overrightarrow{PQ} = 2\hat{i} + 5\hat{j} - 3\hat{k}$

$PM = \left|\frac{\overrightarrow{PQ}.\overrightarrow{PM}}{|\overrightarrow{PM}|}\right| = \left|\frac{4 - 10 - 3}{3}\right| = 3$

3. Ans. 0.5

Sol. lines are $\frac{x - \frac{1}{b}}{c} = \frac{y}{-b} = \frac{z - \frac{1}{b}}{0}$ &

$\frac{x - \frac{2}{b}}{a} = \frac{y}{b} = \frac{z - \frac{3}{b}}{c}$ are perpendicular

$\Rightarrow$ $ac - b^2 = 0 \Rightarrow b^2 = ac$

Now $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

$= \frac{1}{2}\left(\frac{a}{c} + \frac{c}{a}\right) - \frac{1}{2} \geq \frac{1}{2}(2) - \frac{1}{2}$

$\left(\because x + \frac{1}{x} \geq 2, x > 0\right)$

$\Rightarrow$ minimum value of $\cos B = \frac{1}{2}$ exist when $a = c$

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4. Ans. 1.00

Sol. $\hat{a}+2\hat{b}+\hat{c}=\hat{a}\times\hat{c}$ ....(i)

Taking dot with $\hat{a}$ we get

$1+2\hat{a}.\hat{b}+\hat{a}.\hat{c}=0$

Let $\hat{a}.\hat{c}=x$

$\Rightarrow 2\hat{a}.\hat{b}=-1-x$

From (i), $\hat{a}+2\hat{b}=-\hat{c}+\hat{a}\times\hat{c}$

Squaring $1+4+4\hat{a}.\hat{b}=1+1-(\hat{a}.\hat{c})^2$

$\Rightarrow 5-2-2x=2-x^2 \Rightarrow x=1$

$\Rightarrow \hat{a}.\hat{c}=1 \Rightarrow \hat{a}=\hat{c} \Rightarrow$ From (i)

$\hat{a}=-\hat{b}$

$\Rightarrow |\hat{a}\times\hat{b}+\hat{b}\times\hat{c}+\hat{a}+\hat{b}+\hat{c}|$

$|(\hat{a}\times-\hat{a})+(-\hat{a}\times\hat{a})+\hat{a}-\hat{a}+\hat{c}|=|\hat{c}|=1$

5. Ans. 3.12 or 3.13

Sol. $\vec{b}.(\vec{b}\times\vec{\alpha}+\vec{a})=\vec{b}.\vec{\alpha} \Rightarrow \vec{b}.\vec{a}=\vec{b}.\vec{\alpha}$ ...(1)

$\because$ proejction of $\vec{\alpha}$ on $\vec{b}=\vec{\alpha}.\hat{b}$

$\Rightarrow \vec{\alpha}.\hat{b}=3 \Rightarrow \vec{\alpha}.\vec{b}=3 \left(\because |\vec{b}|=1\right)$

$\therefore \vec{b}.\vec{a}=3$ (from (1))

Now $\vec{b}\times(\vec{b}\times\vec{\alpha}+\vec{a})=\vec{b}\times\vec{\alpha}$

$\Rightarrow (\vec{b}.\vec{\alpha})\vec{b}-b^2\vec{\alpha}+\vec{b}\times\vec{a}=\vec{b}\times\vec{\alpha}$

$\Rightarrow 3\vec{b}-\vec{\alpha}+\vec{b}\times\vec{a}=\vec{b}\times\vec{\alpha}$

$\Rightarrow 3\vec{b}-\vec{\alpha}+\vec{b}\times\vec{a}=\vec{\alpha}-\vec{a}$

$\Rightarrow \vec{\alpha}=\frac{1}{2}\left(3\vec{b}+\vec{a}+\vec{b}\times\vec{a}\right)$

Projection of $\vec{\alpha}$ along $\vec{a}$

$=\vec{\alpha}.\hat{a}=\vec{\alpha}.\frac{\vec{a}}{4}=\frac{1}{8}\left[3\vec{b}+\vec{a}+\vec{b}\times\vec{a}\right].\vec{a}$

$=\frac{1}{8}\left[3\vec{b}.\vec{a}+|\vec{a}|^2\right]=\frac{1}{8}[3\times3+16]=\frac{25}{8}=3.125$

SECTION–IV

1. Ans. (A)→(Q,S); (B)→(T); (C)→(P); (D)→(R)

Sol. (A) Let $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$

$c_1^2+c_2^2+c_3^2=2$,

$\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}=\frac{1}{2}=\frac{\vec{a}.\vec{c}}{|\vec{a}||\vec{c}|}=\frac{c_1+c_2}{2} \Rightarrow c_1+c_2=1$

$c_2=1-c_1$

simillarly $c_3=1-c_1$

$\Rightarrow c_1^2+(1-c_1)^2+(1-c_1)^2=2 \Rightarrow c_1=0,\frac{4}{3}$

(B) Let $\vec{r}=r_1\hat{i}+r_2\hat{j}+r_3\hat{j}$

$\vec{r}\times(2\hat{i}+\hat{j}+\hat{k})=\hat{j}-\hat{k}$

$\Rightarrow (r_2-r_3)\hat{i}-\hat{j}(r_1-2r_3)+\hat{k}(r_1-2r_2)=\hat{j}-\hat{k}$

comparing corresponding components

$r_2=r_3,\ r_1-2r_3=-1,\ r_1-2r_2=-1$

$\Rightarrow r_1-2r_2=-1$ There are infinite solutions.

Let $r_2=\lambda=r_3$ $\because r_1=2r_2-1$ $\therefore r_1=2\lambda-1$

$\vec{r}=-\hat{i}+\lambda(2\hat{i}+\hat{j}+\hat{k})$

$\Rightarrow \vec{r}=3\hat{i}+2\hat{j}+2\hat{k}$ for $\lambda=2$

(C) Let equation of plane be $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

it passs through $(3,4,5) \Rightarrow \frac{3}{a}+\frac{4}{b}+\frac{5}{c}=1$ ...(1)

and A(a,0,0), B(0,b,0), C(0,0,c)

equation of plane throught A and parallel to yz plane is x = a

equation of plane throught B and parallel to zx plane is y = b

equation of plane throught C and parallel to xy plane is z = c

$\Rightarrow$ point of intersection of three planes is (a,b,c)

a = 6, b = 16, c = 20 satisfy ...(1)

(D) $(\vec{r}-\vec{a})\times\vec{b}=0$

$\Rightarrow \vec{r}=\vec{a}+t\vec{b}$

$=(\hat{i}+\hat{j})+t(2\hat{i}-\hat{k})$

$=(1+2t)\hat{i}+\hat{j}-t\hat{k}$

simillarly for second line

$\vec{r}=\vec{b}+s\vec{a}$

$\vec{r}=(2+s)\hat{i}+s\hat{j}-\hat{k}$

if two lines intersect

$1+2t=2+s$

$1=s$

$t=1$

$\Rightarrow$ Point of intersection is (3,1,–1)

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